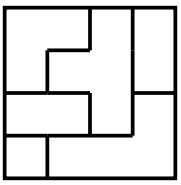
Math Kangaroo Practice - Grades 3-4

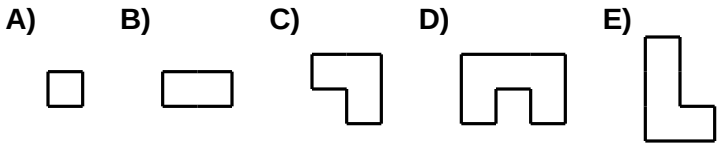
Selected problems from 2002 (#4, #7, #9) and 2003 (#9, #12, #14)
Each original problem is followed by three similar practice problems.

Set 1 - Cutting a square into pieces

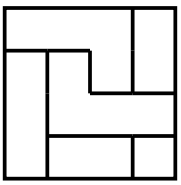
Based on 2002, Problem 4

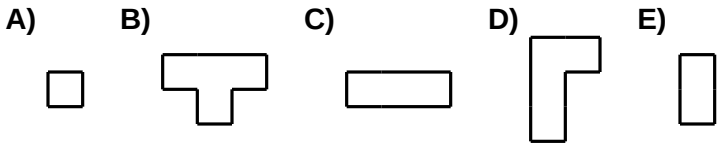
Original 2002 #4. The square was cut along the lines. Which of the shapes below was **not** obtained this way?



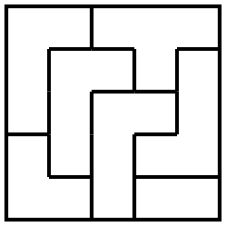


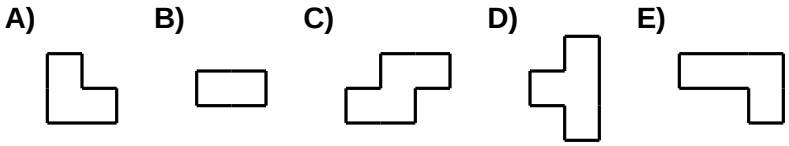
Practice 1.1 The square was cut along the lines. Which of the shapes below was **not** obtained this way?



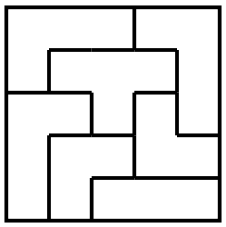


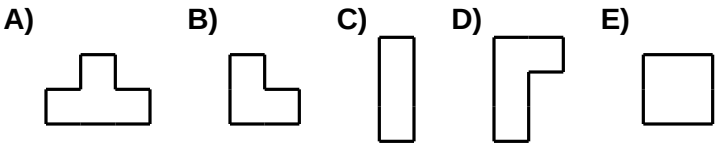
Practice 1.2 This square was cut along the lines. Which shape was **not** obtained?





Practice 1.3 This square was cut along the lines. Which shape was **not** obtained?

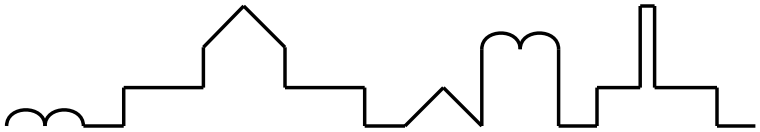


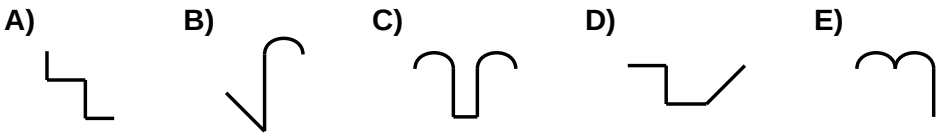


Set 2 - Pieces of a skyline

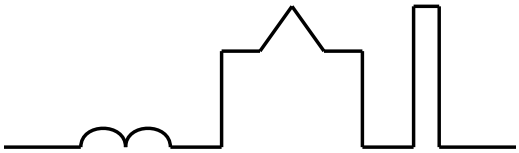
Based on 2002, Problem 7

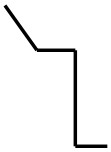
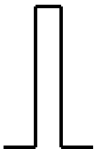
Original 2002 #7. Far away we see the skyline of a castle. Which of the pieces **cannot** belong to the skyline?



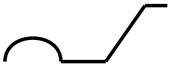
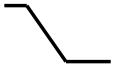


Practice 2.1 Here is the skyline of a town. Which piece **cannot** belong to it?

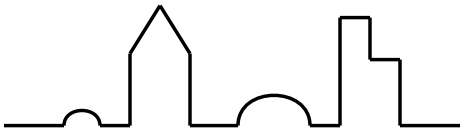


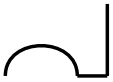
A)
B) 
C)
D)
E) 

Practice 2.2 Here is the skyline of a castle. Which piece **cannot** belong to it?

A)
B) 
C) 
D)
E)

Practice 2.3 Here is a skyline with a small arch, a tower and a dome. Which piece **cannot** belong to it?

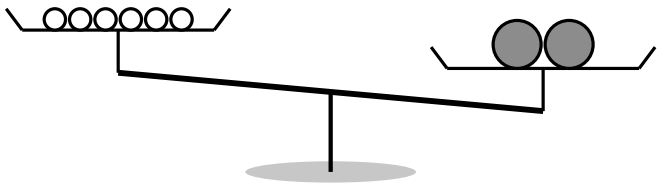


A) 
B)
C)
D)
E)

Set 3 - Balance scales

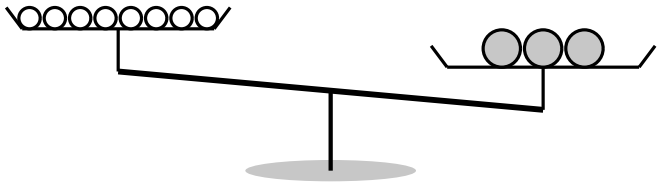
Based on 2002, Problem 9

Original 2002 #9. On one plate of a balance there are 6 oranges (small circles) and on the other there are melons (big circles). When we put one more melon, exactly like the others, on the orange plate, the balance is level. The weight of one melon is the same as:



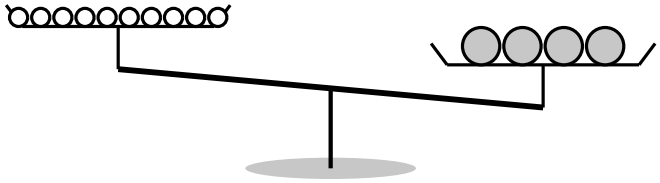
A) 2 oranges **B)** 3 oranges **C)** 4 oranges
D) 5 oranges **E)** 6 oranges

Practice 3.1 On one plate there are 8 plums (small circles) and on the other 3 lemons (gray circles). When we put one more lemon on the plum plate, the balance is level. One lemon weighs the same as:



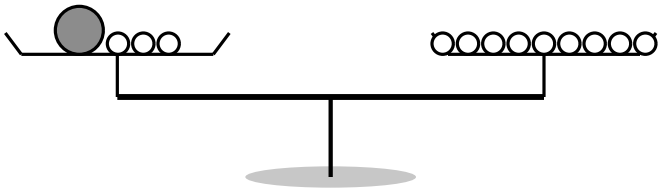
A) 2 plums **B)** 3 plums **C)** 4 plums
D) 5 plums **E)** 8 plums

Practice 3.2 On one plate there are 10 cherries (small circles) and on the other 4 apples (gray circles). When we put 2 more apples on the cherry plate, the balance is level. One apple weighs the same as:



A) 2 cherries **B)** 3 cherries **C)** 4 cherries
D) 5 cherries **E)** 10 cherries

Practice 3.3 The balance is level. On the left plate there is 1 melon (big circle) and 3 oranges; on the right plate there are 9 oranges. How many oranges weigh the same as **2 melons**?

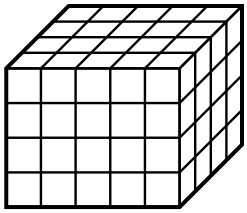


A) 6 **B)** 8 **C)** 9 **D)** 12 **E)** 18

Set 4 - Red outside, blue inside

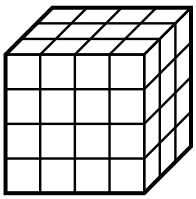
Based on 2003, Problem 9

Original 2003 #9. Chris built the brick in the picture using red and blue cubes of the same size. The outside of the brick is completely red, but all cubes used inside are blue. How many blue cubes did Chris use?



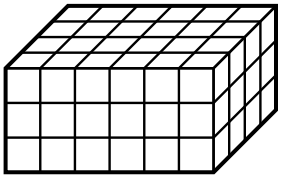
A) 12 **B)** 24 **C)** 36 **D)** 40 **E)** 48

Practice 4.1 Maya built this brick (4 cubes wide, 4 cubes tall, 3 cubes deep). The outside is completely red and all the cubes inside are blue. How many blue cubes did she use?



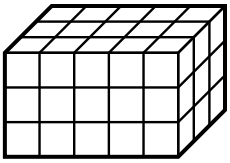
A) 2 **B)** 4 **C)** 6 **D)** 8 **E)** 12

Practice 4.2 Leo built this brick (6 cubes wide, 3 cubes tall, 4 cubes deep). The outside is completely red and all the cubes inside are blue. How many blue cubes did he use?



A) 4 **B)** 6 **C)** 8 **D)** 12 **E)** 16

Practice 4.3 Anna built this brick (5 cubes wide, 3 cubes tall, 3 cubes deep). The outside is completely red and the cubes inside are blue. How many **red** cubes did she use?

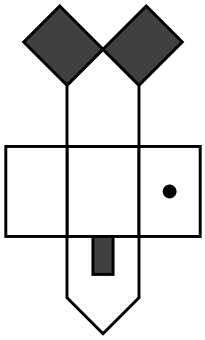


A) 30 **B)** 36 **C)** 39 **D)** 42 **E)** 45

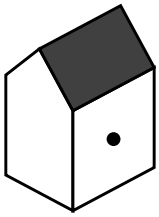
Set 5 - Folding a house

Based on 2003, Problem 12

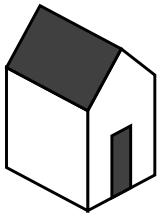
Original 2003 #12. The picture has been drawn on paper and cut out to make a house. Which of the houses did it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*



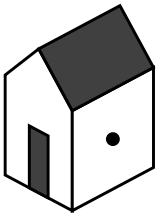
A)



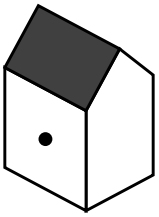
B)



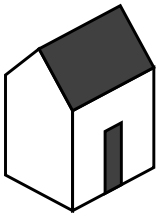
C)



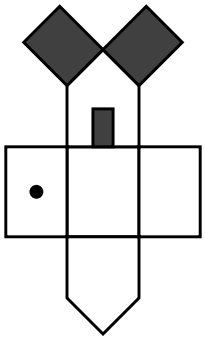
D)

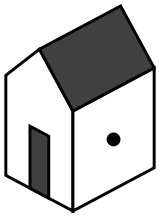


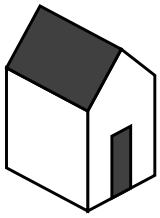
E)

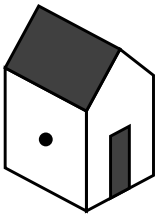


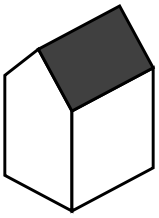
Practice 5.1 This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

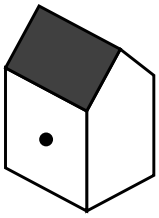


A) 

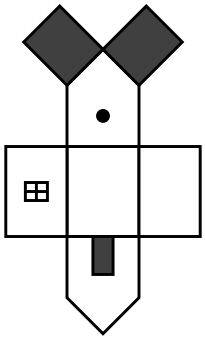
B) 

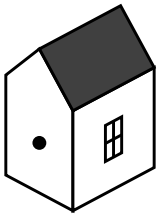
C) 

D) 

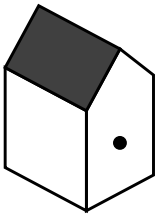
E) 

Practice 5.2 This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

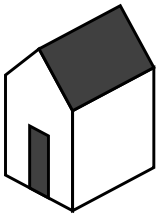


A) 

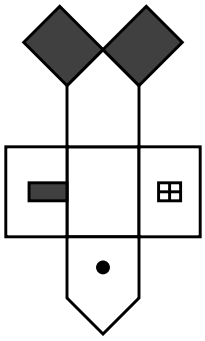
B)

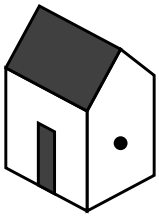
C) 

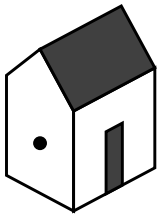
D)

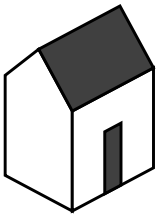
E) 

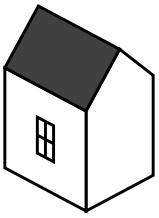
Practice 5.3 This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

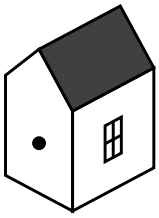


A) 

B) 

C) 

D) 

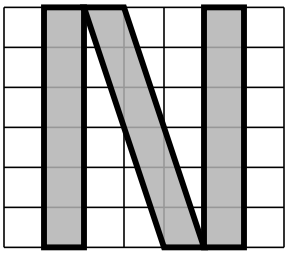
E) 

Set 6 - Area of a letter on squared paper

Based on 2003, Problem 14

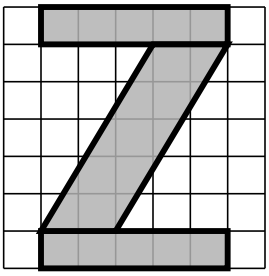
Original 2003 #14. If the side of each little square is 1 cm, what is the area of the letter N (in square cm)?

A) 14 **B)** 15 **C)** 16 **D)** 17 **E)** 18



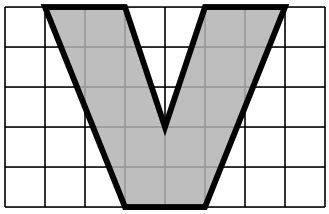
Practice 6.1 If the side of each little square is 1 cm, what is the area of the letter Z (in square cm)?

A) 16 **B)** 18 **C)** 20 **D)** 22 **E)** 24

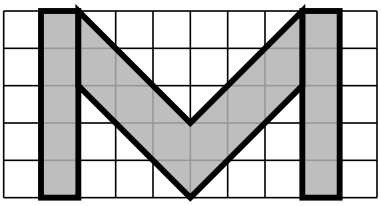


Practice 6.2 If the side of each little square is 1 cm, what is the area of the letter V (in square cm)?

A) 15 **B)** 16 **C)** 17 **D)** 18 **E)** 20



Practice 6.3 If the side of each little square is 1 cm, what is the area of the letter M (in square cm)?



A) 18 **B)** 20 **C)** 22 **D)** 24 **E)** 26